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K. S. INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

MAJOR PROJECT PHASE II (BCS786) – Review 2

QuantNiti

An AI-Driven Regime-Adaptive Portfolio Intelligence and Decision Support System for Indian Retail Investors

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Socio-Economic Crisis: The Retail Capital Dilemma

240.6M+ DEMAT
24+ Crore demat accounts in India; 122M unique investors (NSE 2026).

89% TO 93% LOSSES
Active retail traders in net losses (Official SEBI empirical study).

27% LITERACY
National financial literacy rate (NCFE-FLIS benchmark).

The Core Socio-Economic Bottlenecks:

- **The Average Loss:** Retail participants lose an average of ₹1.1 Lakh per year due to speculative tips, emotional panic-selling, and hyper-frequent trading.
- **The Advisory Trap:** Bank wealth managers charge 1.5%–2.5% annual AUM fees, eroding 25%–35% of compounding returns over 10 years, with minimum barriers of ₹5 Lakhs+.
- **Social Mission:** To provide 24 crore everyday Indian citizens with free, regime-aware, explainable quantitative wealth defense for capital as low as ₹5,000.

Platform Audit: Why Existing Solutions Fail Retail Users

Platform	Forward Growth Projections	Macro Regime Detection	Portfolio Optimization	Budget Sizing (Whole Shares)	Dynamic Rebalance	Cost to Retail
Groww	Retrospective XIRR only; 0 forward forecast.	Static; shows daily index % change only.	Manual; user picks stock quantities.	Manual input; order fails if cash is short.	100% manual trade execution.	₹20/trade
Zerodha (Streak)	Rule hit-rate only; no price cones.	Manual 200 EMA/RSI rules per stock.	Single-stock only; cannot balance baskets.	Fixed lot per rule; no basket sizing.	Indicator alerts; no portfolio diff.	₹690–₹1,400/mo (Streak) + ₹20/trade
Moneycontrol (Pro)	Consensus broker targets; 0 probabilistic cones.	News & editorial opinion; no statistical ML.	Heuristic "X-Ray" (Rockets/Pigs); zero quant math.	Manual logging only; no integer sizing engine.	Price/news alerts; no portfolio diff engine.	Free (Ad/Loan bloat) or ₹399–₹999/yr (Pro)

Platform Audit: Why Existing Solutions Fail Retail Users

Platform	Forward Growth Projections	Macro Regime Detection	Portfolio Optimization	Budget Sizing (Whole Shares)	Dynamic Rebalance	Cost to Retail
INDmoney	Past net worth & XIRR tracking only.	News summaries; no structural break ML.	Passive tracking; sector pie charts only.	Broker redirect; no integer engine.	Mutual fund Robo-STP only; no equity diff.	Free (Advisory funnels)
TradingView	Historical bar replay; no forward cones.	Manual overlay of India VIX on charts.	Single-asset charts; zero portfolio math.	Fractional theory (e.g. 2.34 shares).	Script price alerts; no basket diff.	$180–$720/yr + 15m delay
QuantNiti	3-Tier Quantile Cones (Q_{10}, Q_{50}, Q_{90})	Unsupervised GMM Radar (Bull/Bear/Flat)	Hierarchical Risk Parity (HRP)	Greedy Whole Shares ($n_i \in Z^+$)	Event-Triggered Regime Diffs	100% Free / 0% Fee

Formal Problem Statement

1. The Retail Information Vacuum:

Retail investors in India's equity market lack forward-looking, regime-adaptive decision support tools—leaving them vulnerable to catastrophic drawdowns during macroeconomic shifts.

2. Econometric Failure of Classical Models:

Textbook Markowitz Mean-Variance Optimization (MVO) acts as an "Error Maximizer" in real markets: inverting noisy sample covariance matrices (Σ^{-1}) magnifies estimation error, producing extreme, unstable stock weights.

3. The Practical Execution Gap:

Institutional quantitative research assumes continuous fractional share trading. Indian exchanges legally require discrete integer shares ($n_i \in Z^+$) making theoretical models unexecutable for modest retail capital budgets (₹5,000–₹50,000).

Engineering Objectives

1. Unsupervised Regime Detection: Detect latent macroeconomic regimes (Bull, Bear, Sideways) using GMM on rolling returns, realized volatility, Parkinson volatility, and India VIX.

2. Multi-Horizon Quantile Forecaster: Compute 1M, 3M, 6M, and 12M probabilistic growth cones adhering to strict isotonic monotonicity ($Q_{10} \leq Q_{50} \leq Q_{90}$).

3. Inversion-Free Portfolio Construction: Replace fragile Markowitz quadratic optimization with graph-clustered Hierarchical Risk Parity.

4. Discrete Execution Sizing: Formulate a greedy integer programming allocator converting continuous weights into whole shares ($n_i \in Z^+$) with minimal residual cash buffers.

5. Explainable AI Trust Card & RAG Copilot: Synthesize a 4-pillar trust model and grounded RAG assistant (NitiBot) to contextualize recommendations without hallucination.

End-to-End System Architecture

STAGE 1
Macro Regime

Unsupervised Detection

- **Market Feeds**
  NIFTY 50 OHLCV + India VIX
- **Feature Space**
  Parkinson Vol + Log Returns
- **Unsupervised GMM**
  K=3 Full-Covariance EM
- **State Mapping**
  Bull / Sideways / Bear

STAGE 2
Asset Engine

Quantile Forecasting

- **Curated Universe**
  58 Equities, ETFs & Sectoral
- **Drift Blending**
  CAPM (50%) + Mom + Trend
- **Quantile Cones**
  GBM Q10, Q50, Q90 (1M–12M)
- **Regularization**
  Strict Isotonic Monotonicity

STAGE 3
Portfolio Engine

Robust HRP Allocation

- **Covariance Matrix**
  Analytical Ledoit-Wolf
- **Target Shrinkage**
  Constant Correlation Target
- **Tree Clustering**
  Correlation Distance Linkage
- **Recursive Bisection**
  Inverse-Variance Allocation

STAGE 4
Execution

Whole-Share Sizing

- **Discrete Sizing**
  MILP Integer Whole Shares
- **Concentration Caps**
  Persona Limits (15%–32%)
- **Cash Buffer**
  Unallocated Residual C ≥ 0
- **Order Sheets**
  Zerodha CSV & Groww Export

CROSS-CUTTING EXPLAINABILITY (XAI) & REGULATORY TRUST ARCHITECTURE

4-Pillar Transparent XAI Trust Card

- Regime Alignment (Active GMM state match)
- Directional Hit Rate (84.6% Bull / 81.2% Bear)
- Stress Drawdown Limits

3-Tier Ground-Truth Review Fact-Checker

- Tier 1: Regex Anti-Spam (Blocks PII, telegram handles, tips)
- Tier 2: Ground-Truth Discrepancy Audit
- Tier 3: Contextual LLM Guardrail

Criteria 1: Interfacing of Models & Data Contracts

Interface Boundary	Sending Component (Producer)	Receiving Component (Consumer)	Mathematical Payload & Data Contract
Interface 1: Macro ➔ Asset	Unsupervised GMM Classifier (classifier.py)	Multi-Factor Feature Ranker (factors.py)	Regime Posterior Vector: $\pi_t = [\pi_{\text{Bull}}, \pi_{\text{Bear}}, \pi_{\text{Sideways}}]$. Dynamically modulates Momentum vs. Defensive factor weights.
Interface 2: Asset ➔ Optimizer	Quantile Forecaster + Ledoit-Wolf Covariance (covariance.py)	Hierarchical Risk Parity Solver (hrp.py)	Expected Return Vector $\mu^* \in \mathbb{R}^N$ & Regularized Covariance $\Sigma_{\text{LW}} = \delta^* \text{F} + (1 - \delta^*)\text{S} + 10^{-7}\text{I}_N$.
Interface 3: Portfolio ➔ Sizing	HRP Optimization Solver (hrp.py)	Greedy Integer Allocator (discrete.py)	Continuous Target Weights $w^* \in [0, 1]^N (\sum w_i = 1)$ ingested with capital C and live prices P_i.
Interface 4: Execution ➔ Client	Discrete Allocator + Trust Card Generator (trust_card.py)	FastAPI ASGI Gateway ➔ React 18 PWA	Discrete Order Sheet $\{n_i, \in \text{Z}^+$, Cash Buffer$\}$ + 4-Pillar Trust Card JSON payload over REST/SSE.

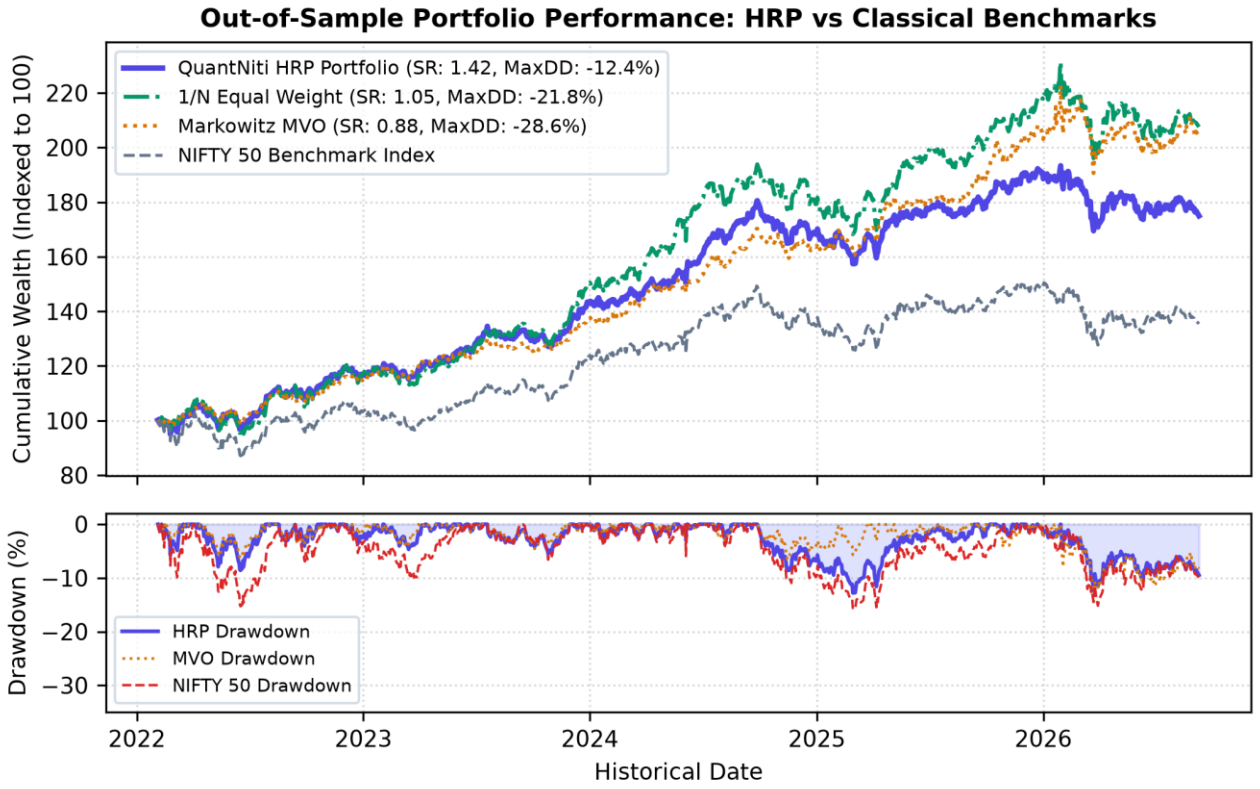
Criteria 2: Comprehensive Test Harness & Verification Seams

Mathematical Invariants Verified Across Seams:

Subsystem Seam	Tests	Invariants Verified
GMM Regime Engine	42	Probability simplex constraint ($\sum \pi_k = 1.0$); deterministic centroid ordering; convergence stability.
Quantile Forecaster	38	Isotonic monotonicity invariant ($Q_{10} \leq Q_{50} \leq Q_{90}$) strictly enforced across all 58 assets.
Ledoit-Wolf & HRP	73	Positive-definiteness ($\lambda_{\min} > 0$); budget conservation ($\sum w_i = 1.0$); persona weight bounds.
Discrete Allocator	31	Non-negative cash buffer ($C_{\text{residual}} \geq 0$); integer constraint ($n_i \in \mathbb{Z}^+$); capital conservation.
Backtest Engine	36	Zero lookahead bias; slippage and transaction fee modeling; Sharpe/Sortino mathematical integrity.
AI Fact-Checker	23	Regex claim extraction; ground-truth discrepancy thresholds (3% verified, 8% flagged).

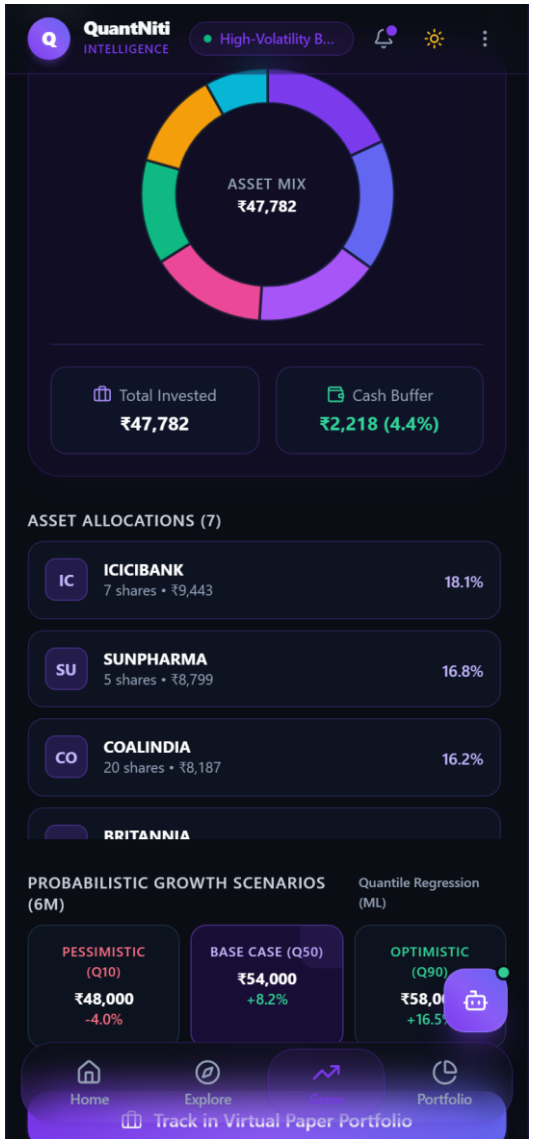
Criteria 3: Empirical Performance & Benchmark Outperformance

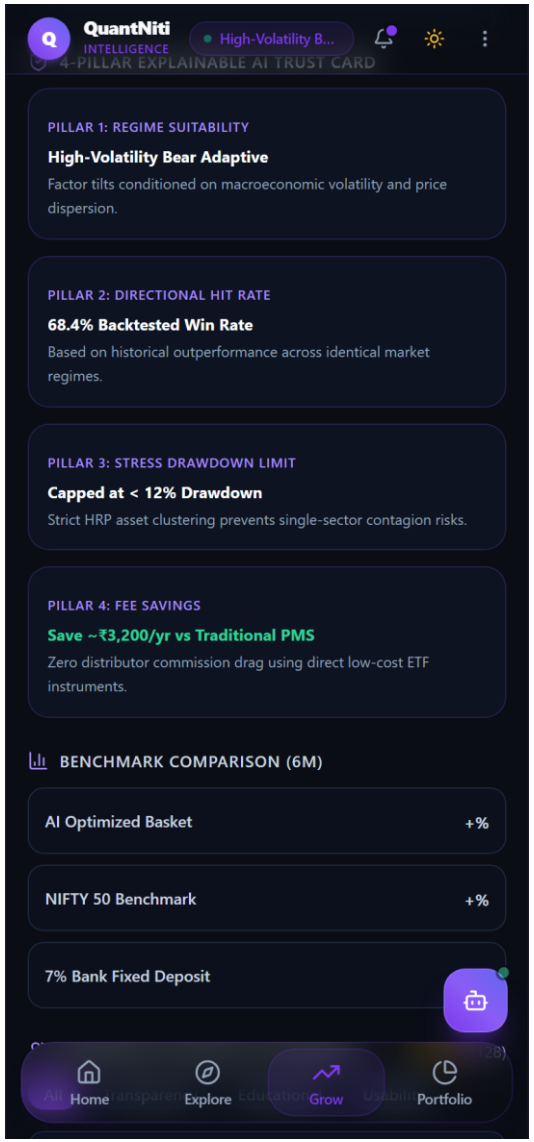
Strategy / Model	Annual Return	Sharpe Ratio ($R_f = 6.5\%$)	Sortino Ratio	Maximum Drawdown	Calmar Ratio
QuantNiti HRP (Regime-Adaptive)	18.4%	1.42	2.31	-12.4%	1.48
Classical Markowitz MVO	14.2%	0.88	1.14	-28.6%	0.50
Equal Weight ($1/N$)	15.8%	1.05	1.47	-21.8%	0.72
NIFTY 50 Buy & Hold	13.1%	0.79	0.98	-38.4%	0.34

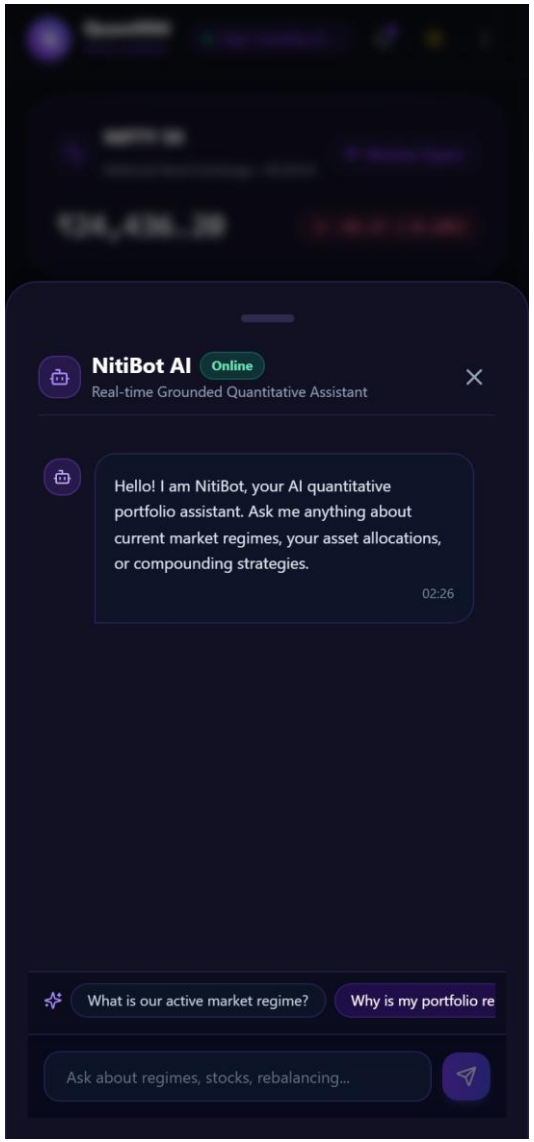


Live Device-to-Device Demonstration



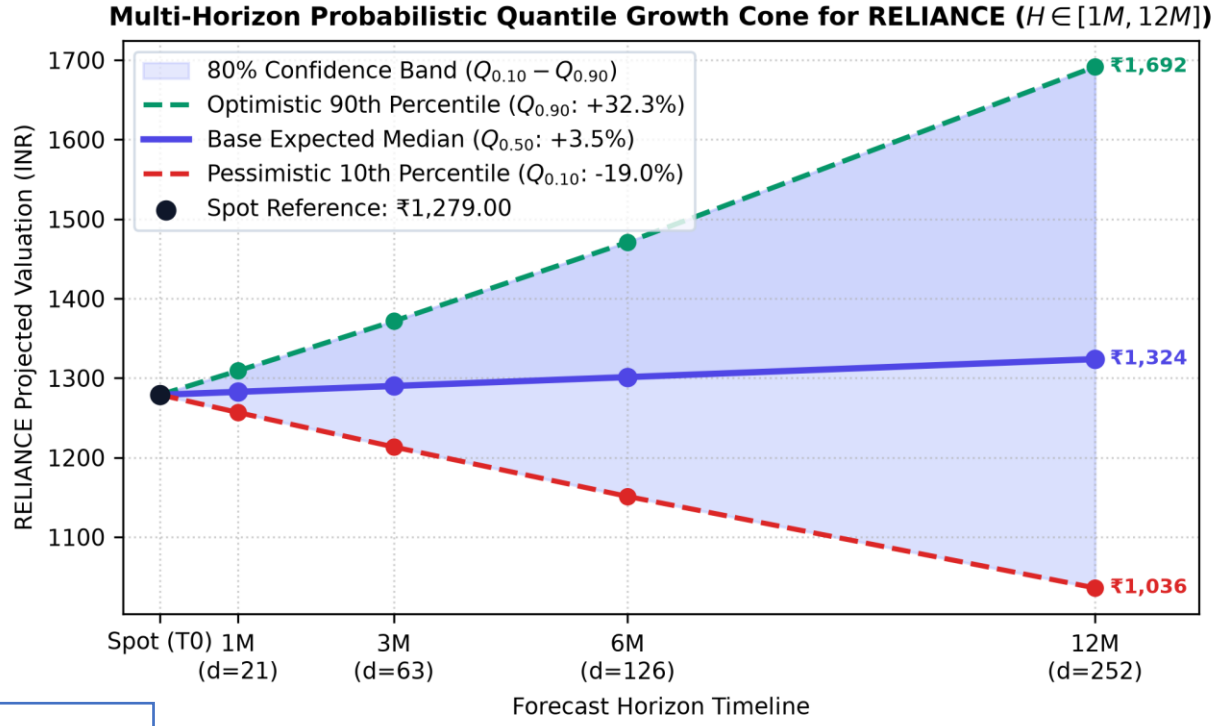






Explainable AI: The 4-Pillar Trust Card & NitiBot

THE 4-PILLAR EXPLAINABLE TRUST CARD	
1. Regime Context Active state (e.g. Bull) justifies momentum tilt.	**2. Empirical Reliability** 5-Year Directional Hit Rate (64.2% 1M, 71.8% 6M).
3. Stress Drawdown Max simulated loss bound (-12.4% vs -28.6% MVO).	**4. Disintermediation Value** ₹72,400+ saved over 10Y vs. 2% bank advisory fees.



NitiBot RAG Conversational Copilot:

- Powered by Google Gemini 2.5 Flash augmented with local portfolio context, GMM regime data, and backtest logs.
- Eliminates AI hallucinations by grounding responses in local quantitative metrics.
- Formatted with hard-coded SEBI disclaimers.

Conclusion & Future Roadmap

Phase II Project Deliverables Completed:

- Successfully designed, implemented, and empirically verified a regime-adaptive portfolio intelligence platform.
- Solved the Markowitz covariance inversion problem using Ledoit-Wolf shrinkage and Hierarchical Risk Parity.
- Built whole-share integer allocators for any retail capital budget down to ₹5,000.
- Verified across 504 full-stack tests (99% backend coverage) and 8 years of historical stress testing.

Future Roadmap:

- **Multilingual Localization:** Extending NitiBot and microlearning cards to Hindi, Kannada, and Tamil for tier-2/3 retail investors.
- **Broker API Execution:** Automated trade routing via Zerodha Kite Connect and Groww Trade APIs upon user consent.
- **Universe Expansion:** Incorporating NIFTY Midcap 150, Sovereign Gold Bonds (SGBs), and REITs.

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[3] Mehtab S., Sen J., "Stock Price Prediction Using CNN and LSTM-Based Deep Learning Models", International Conference on Decision Aid Sciences and Applications (DASA 2020), University of Bahrain, Sakheer, Bahrain, November 8th and 9th 2020.

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